

TANVI GANESH JOSHI

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EDUCATION

University of California, San Diego

MS in Computer Science and Engineering – GPA: 3.97/4.0

San Diego, CA

Sept 2024 – Jun 2026

Indian Institute of Technology, Bombay

B.Tech. in Mechanical Engineering, Minor in CSE – CPI: 8.85/10.0

Mumbai, India

Nov 2020 – May 2024

EXPERIENCE

BeOne Medicines

June 2025 – Mar 2026

Advanced Analytics & ML Intern

San Carlos, CA

- Built production pipeline to score **10,000+ HCPs** from **1M+ biomedical interactions** ingested across **20+ data sources**.
- Redesigned KOL influence network incorporating Spoke-Hub referral dynamics, distinguishing local vs. regional/national KOL tiers; deployed unified entity model across **multiple oncology products**, scalable to full portfolio.
- Enriched social media data model via Meltwater API integration; automated Dataiku workflows and coverage audits, reducing manual analytics effort by **~70%** and preventing redundant third-party data purchases.

Halicioğlu Data Science Institute, UCSD

Apr 2026 – Present

Teaching Assistant – DSC102: Systems for Scalable Analytics

San Diego, CA

- TA for **244 students**; topics include distributed computing, MapReduce/Hadoop, Spark, and end-to-end ML at scale.
- Led weekly discussions, lecture reinforcement, and exam prep; supported **4 programming assignments** on AWS and Spark.
- Evaluated student submissions over large datasets; supported topics including dataflow systems, and scalable AI workloads.

American Express

May 2023 – July 2023

Data Scientist, Decision Science & Strategy Intern

Mumbai, India

- Built and productionized end-to-end propensity model over **12M+ cardmember records** to predict B2B merchant transactions and select ad-targetable cohorts; engineered rule-based filters for high-precision training labels.
- Designed taxonomy mapping merchants, constructing ground-truth labels from raw transaction logs for supervised training.
- Implemented ML training pipeline with feature aggregation, normalization, PCA, and stratified splits; achieved **~0.80 ROC-AUC** / **~0.70 F1**, producing deployable scored cohorts with **~2.5× lift**.

PROJECTS

Multi-Agent LLM System with RL-Based Belief Inference (Avalon) | Python, PyTorch

2026

- Built multi-agent LLM simulation with role-conditioned prompting, structured memory, and RL-based belief inference (suspicion + role prediction); supported by game engine with partial-information state tracking and APIs.
- Designed game loop enforcing partial-information constraints with discussion, voting, and team selection interfaces.
- Improved accusation accuracy to **0.73** and achieved **0.83** accuracy for low-information agents via RL policy learning.

Two-Tower Neural Recommendation System | PyTorch, FAISS, Python

2026

- Built two-stage neural retrieval & ranking pipeline on MovieLens 1M; dual-encoder with InfoNCE loss and **FAISS IVFFlat** indexing achieving **5×** speedup and **0.006ms/query** latency.
- Ranking MLP trained on hard negatives improved **NDCG@10** by **110%** and **MRR** by **85%** over retrieval-only baseline; applied temporal per-user splits and evaluated with Recall@K, NDCG@K, and MRR.

Customer Churn Prediction with Causal Attribution | XGBoost, SHAP, scikit-learn

2025

- Built XGBoost churn classifier on IBM Telco dataset (7K customers); engineered 5 domain features; achieved **AUC-ROC 0.849**, **68% precision** and **2.55× lift** at top-20% risk tier.
- Segmented churned population into 4 interpretable cohorts via K-Means in SHAP space; top 2 cohorts captured 75% of churners and **\$257K** of **\$327K** revenue at risk, enabling targeted retention campaigns projected to cut churn 18–22%.

Autonomous Driving Trajectory Prediction | PyTorch, Python

2025

- Built Social-LSTM trajectory predictor for scenes with up to **50 interacting driving agents**; applied ego-centric normalization, rotation/mirroring augmentation, and physics-aware loss (MSE + direction + curvature).
- Reduced displacement MSE from **24.0→7.7 (68% reduction)** via LR scheduling and early stopping.

Speaker Attribution using Sequence Models | PyTorch

2024

- Built Transformer encoder-decoder with multi-head attention, masking, residual connections, and layer norm; trained speech segment classifier (**~85% accuracy**) and autoregressive LM (**~85 perplexity**) with causal masking.
- Tested AliBi positional encodings and analyzed attention maps to study long-range dependency learning.

TECHNICAL SKILLS

Languages: Python, SQL, HTML, $\text{\LaTeX}$

Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, HuggingFace Transformers, sentence-transformers, LangChain

Data & Systems: Hive, Hadoop, Spark, Airflow, Dataiku, Ray, AWS (S3, EC2), FAISS, REST APIs

Focus Areas: NLP & LLMs, Ranking & Targeting, Predictive Modeling, Recommender Systems, Agentic AI, ML & DL

Miscellaneous: KVPY Fellow, ME119 TA (IITB), Dept Academic Mentor (IITB), Grad Student Mentor (UCSD, IITB)